

Bitwise differential trail through **Blink-128** found by SAT

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July 25, 2026

1 Blink-128

Maximal differential probability: 2^{-14}

0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	8	
0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	8	
BN																													
0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	8	
0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	8	
Fullband																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	
Fullband																													
0	0	2	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	2	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Fullband																													
0	0	0	0	0	0	0	0	0	8	0	0	8	0	0	8	0	8	0	0	8	0	8	0	0	8	8	0	0	8
0	0	0	0	0	0	0	0	0	8	0	0	8	0	0	8	0	8	0	0	8	0	8	0	0	8	8	0	0	8

2 FwdRound

Maximal differential probability: $2^{-6.0}$

0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	8
0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	8
SubCells																												
0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2	
0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2	
MixColumns																												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	
RK																												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	
RK																												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	
Permutation																												
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	

3 SubCells

Maximal differential probability: $2^{-6.0}$

0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	8
0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	8
∂B_{102}	∂B_{103}	∂B_{104}	∂B_{105}	∂B_{106}	∂B_{107}	∂B_{108}	∂B_{109}	∂B_{110}	∂B_{111}	∂B_{112}	∂B_{113}	∂B_{114}	∂B_{115}	∂B_{116}	∂B_{117}	∂B_{118}	∂B_{119}	∂B_{120}	∂B_{121}	∂B_{122}	∂B_{123}	∂B_{124}	∂B_{125}	∂B_{126}	∂B_{127}	∂B_{128}	
0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2
0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2
0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2

4 MixColumns

Maximal differential probability: $2^{-0.0}$

0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2
0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	2
MixColumns																											
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0

5 FwdRound

Maximal differential probability: $2^{-2.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
SubCells																													
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0
MixColumns																													
2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
RK																													
2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
RK																													
2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
Permutation																													
0	0	2	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	2	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

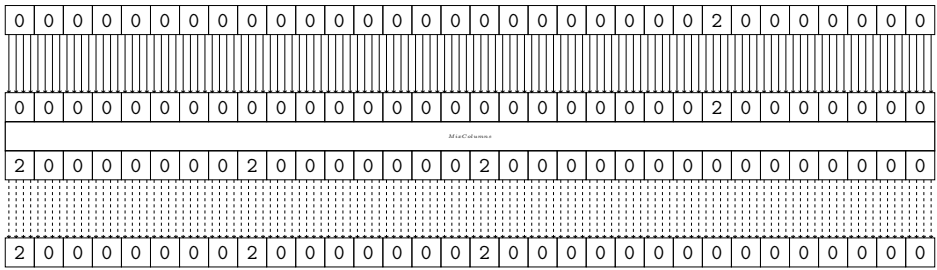
6 SubCells

Maximal differential probability: $2^{-2.0}$

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0
∂B_{102}	∂B_{103}	∂B_{104}	∂B_{105}	∂B_{106}	∂B_{107}	∂B_{108}	∂B_{109}	∂B_{110}	∂B_{111}	∂B_{112}	∂B_{113}	∂B_{114}	∂B_{115}	∂B_{116}	∂B_{117}	∂B_{118}	∂B_{119}	∂B_{120}	∂B_{121}	∂B_{122}	∂B_{123}	∂B_{124}	∂B_{125}	∂B_{126}	∂B_{127}	∂B_{128}		
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	

7 MixColumns

Maximal differential probability: $2^{-0.0}$



8 FwdRound

Maximal differential probability: $2^{-6.0}$

0	0	2	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0	0	2	0	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
SubCells																															
0	0	8	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
0	0	8	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
MixColumns																															
0	0	0	0	0	0	0	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	
0	0	0	0	0	0	0	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	
RK																															
0	0	0	0	0	0	0	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	
0	0	0	0	0	0	0	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	
RK																															
0	0	0	0	0	0	0	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	
0	0	0	0	0	0	0	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	
Permutation																															
0	0	0	0	0	0	0	0	0	8	0	0	8	0	0	0	8	0	0	8	0	8	0	0	0	0	8	8	8	0	0	8
0	0	0	0	0	0	0	0	0	8	0	0	8	0	0	0	8	0	0	8	0	8	0	0	0	0	8	8	8	0	0	8

9 SubCells

Maximal differential probability: $2^{-6.0}$

[illegible]

10 MixColumns

Maximal differential probability: $2^{-0.0}$

0	0	8	0	8	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	8	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	8	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
MidColumns																															
0	0	0	0	0	0	0	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0
0	0	0	0	0	0	0	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0	0	0	8	0	8	0	8	0